Test a detector for both missed and false signals

Task 4.2 / Prepare and draw

Name _____ Team _____ Date _____

01 Set up the investigation

Place a paper target where the switch should trigger. Draw the source, trigger and output. Define what counts as a detection before testing.

Set the trigger location and record how many trials you will run with and without movement.

Condition	Before trials 1 / 2 / 3	After trials 1 / 2 / 3
Movement		
No movement		

02 Plan it on paper

Draw the source, trigger, and output. State what should happen in each condition.

My prediction, chosen feature or design limit

Test a detector for both missed and false signals

Task 4.2 / Test, compare and explain

Name _____ Team _____ Date _____

03 Run the test and record what happened

Assemble the detector and move the simulated trigger the same way for each trial. Record the output. Run three trigger and three no-trigger trials. Change one unreliable contact. Repeat both sets.

Version	True detections / 3	False alarms / 3	Contact change or note
Before			
After			

04 Use the reading

Ch. 25 | Teaching a Machine to Listen | printed pp. 231-236
Complete the assigned checks in the interactive reader. Here, identify a page and one idea that helps explain your result or design decision.

05 Individual explanation

Give the detector’s true detections and false alarms. Which change would improve reliability?

Keep these sheets with the work. Save the individual explanation in the lesson’s Explain step.